MTH 331(A) Dr. James Ashe
Take home quiz Due November 11, 2012 **Name:**

10.1: Approximations with Taylor Polynomials.

HW problems 21-22 and 33-36. *Approximate the given quantities using Taylor polynomials with* $n = 3$, *and compute the absolute error in the approximation assuming the exact value is given by a calculator.*

1. $e^{0.2}$

2. $\ln(1.01)$

3. $\sin(\frac{\pi + .02}{2})$ use center $\frac{\pi}{2}$.

10.2: Find the Interval of Convergence.

HW problems 9-10 and 21-24. *Determine the radius of convergence of the following power series. The test the endpoints to determine the interval of convergence.*

4. $\sum \left(\frac{3x}{2}\right)^k$

5. $\sum \frac{k^2 x^{2k}}{k!}$ (use the Ratio test)

HW problems 21-24. *Find the power series representations centered at* 0 *for the following functions using known power series. Give the interval of convergence for the resulting series.*

6. $f(x) = \dfrac{1}{1-2x}$

7. $f(x) = \dfrac{2x^3}{3-3x}$

11.1: Working with Parametric Equations.

HW problems 7-8. *Plot the parametric points for the given values of* t*, and eliminate the parameter to an equation in terms of* x *and* y*.*

8. $x = -t + 6$, $y = 3t - 3$; for $t = -5, -1, 0, 1, 5$

slides_Calc3/images/empty_graph_-10-10_-20-20.png

Figure 0.1: Use the graph to complete the preceding problem.

HW problems 23-24 and 35-36. *Give a set of parametric equations that describes the curves.*

9. The complete curve $x = y^2 - 2y$.

10. A man jogs clockwise at a constant speed around a circular track of radius 50 yards completing one lap in 2.5 minutes.

HW problems 45-48. *Determine dy/dx in terms of t and evaluate it at the given value of t.*

11. $x = 2t$, $y = t^3$; $t = -1$

11.2: Working with Polar Coordinates.

HW problems 9-11. *Graph the points in the following polar coordinates.*

12. $\left(1.5, \frac{7\pi}{4}\right)$, $\left(-1.5, \frac{7\pi}{4}\right)$, $\left(1.5, -\frac{7\pi}{4}\right)$, $\left(1.5, \frac{7\pi}{4} + 2\pi\right)$, $\left(-1.5, \frac{7\pi}{4} + \pi\right)$

slides_Calc3/images/polar_graph_-2-2a.png

Figure 0.2: Use this graph to plot the polar coordinates

HW problems 15-17. *Express the following polar coordinates in Cartesian coordinates.*

13. $\left(2, \frac{7\pi}{4}\right)$

HW problems 21-24. *Express the following Cartesian coordinates in polar coordinates in at least two different ways.*

14. $\left(4, \sqrt{3}\right)$

HW problems 45. *Mark the points on the polar graph that corresponds to the points shown on the Cartesian graph.*

slides_Calc3/images/graph_cos(t)+sin(2t).png *slides_Calc3/images/polar_graph_cos(t)+sin(2t).png*

Figure 0.3: $r = \cos(\theta) + \sin(2\theta)$

15. Mark the following points: A, B, C, D, E, G, H, I, J

11.3: Finding Derivatives and Integral in Polar Coordinates.

HW problems 5-11. *Find the slope of the line tangent to the following polar curves at the given points.*

16. $1 + \cos\theta$; $\left(1 + \frac{\sqrt{3}}{2}, \frac{\pi}{6}\right)$, $(1, 2\pi)$

HW problems 21-25 (odd). *Find the area of the region.*

17. The region inside the limacon $r = 2 + \cos\theta$.

18. The region inside one leaf of the rose $r = \cos 5\theta$.